

Global total Roman domination in graphs

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A *total Roman dominating function* (TRDF) on a graph G is a function $f : V(G) \rightarrow \{0, 1, 2\}$ satisfying the conditions (i) every vertex u for which $f(u) = 0$ is adjacent at least one vertex v for which $f(v) = 2$ and (ii) the subgraph of G induced by the set of all vertices of positive weight has no isolated vertex. The *weight* of a TRDF is the sum of its function values over all vertices. A total Roman dominating function f is called a *global total Roman dominating function* (GTRDF) if f is also a TRDF of the complement $\overline{G}$ of G . The *global total Roman domination number* of G is the minimum weight of a GTRDF on G . In this paper, we initiate the study of global total Roman domination number and investigate its basic properties. In particular, we relate the global total Roman domination and the total Roman domination and the global Roman domination number.

Keywords: Total Roman dominating function; total Roman domination number; global total Roman dominating function; global total Roman domination number.

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1. Introduction

Throughout this paper, G is a simple graph with vertex set $V(G)$ and edge set $E(G)$ (briefly V, E) such that G and the complement $\overline{G}$ have no isolated vertex. The order $|V|$ of G is denoted by $n = n(G)$. Denote by C_n the *cycle* and by P_n the *path* of order n , respectively. For every vertex $v \in V(G)$, the *open neighborhood* of v is the set $N_G(v) = N(v) = \{u \in V(G) \mid uv \in E(G)\}$ and its *closed neighborhood* is the set $N_G[v] = N[v] = N(v) \cup \{v\}$. The *degree* of a vertex $v \in V$ is $\deg(v) = d(v) = |N(v)|$. The *minimum degree* and the *maximum degree* of a graph G are denoted

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by $\delta = \delta(G)$ and $\Delta = \Delta(G)$, respectively. The *open neighborhood* of a set $S \subseteq V$ is the set $N_G(S) = N(S) = \cup_{v \in S} N(v)$, and the *closed neighborhood* of S is the set $N_G[S] = N[S] = N(S) \cup S$. The *S-external private neighborhood* of a vertex $v \in S$ is defined by $\text{epn}(v, S) = \{w \in V(G) \setminus S \mid N[w] \cap S = \{v\}\}$. The *distance* $d_G(u, v)$ between two vertices u and v in a connected graph G is the length of a shortest uv -path in G . The *diameter* of a graph G , denoted by $\text{diam}(G)$, is the greatest distance between two vertices of G . The *subdivision graph* $S(G)$ is the graph obtained from G by subdividing each edge of G . A *packing* in G is a set of vertices that are pairwise at distance at least 3 apart; that is, if u and v are distinct vertices that belong to a packing S , then $d(u, v) \geq 3$. Equivalently, a set S of vertices of G is a packing in G if the closed neighborhoods of vertices in S are pairwise disjoint. A *leaf* of a tree T is a vertex of degree 1, a *support vertex* is a vertex adjacent to a leaf and a *strong support vertex* is a vertex adjacent to at least two leaves. The *corona* graph $\text{cor}(H)$ of a graph H is the graph obtained from H by attaching a leaf to every vertex of H . For a real-valued function $f : V \rightarrow \mathbb{R}$, the *weight* of f is $\omega(f) = \sum_{v \in V} f(v)$, and for $S \subseteq V$ we define $f(S) = \sum_{v \in S} f(v)$, so $\omega(f) = f(V)$. For terminology and notation not defined here we follow [11].

A subset S of vertices of G is a *total dominating set* if $N(S) = V$. The *total domination number* $\gamma_t(G)$ is the minimum cardinality of a total dominating set of G . A total dominating set S of G is a *global total dominating set* of G if S is a total dominating set of both G and $\overline{G}$. The *global total domination number* $\gamma_{gt}(G)$ of G is the minimum cardinality of a global total dominating set. The global total domination number was introduced by Kulli and Janakiram in [6] and has been studied by several authors [2, 5].

A *Roman dominating function* (RDF) on a graph $G = (V, E)$ is defined in [8, 10] as a function $f : V \rightarrow \{0, 1, 2\}$ satisfying the condition that every vertex v for which $f(v) = 0$ is adjacent to at least one vertex u for which $f(u) = 2$. A Roman dominating function $f : V \rightarrow \{0, 1, 2\}$ can be represented by the ordered triple (V_0, V_1, V_2) (or (V_0^f, V_1^f, V_2^f) to refer f) of V , where $V_i = \{v \in V \mid f(v) = i\}$. In this representation, its weight is $\omega(f) = |V_1| + 2|V_2|$. An RDF f is called a *global Roman dominating function* (GRDF) if f is also an RDF of the complement $\overline{G}$ of G . The *global Roman domination number* of G , denoted by $\gamma_{\text{gR}}(G)$, is the minimum weight of a GRDF on G . A $\gamma_{\text{gR}}(G)$ -function is a GRDF of G with weight $\gamma_{\text{gR}}(G)$. The global Roman domination number has been studied in [4, 9].

A *total Roman dominating function* (TRDF) of a graph G with no isolated vertex, is an RDF f on G with the additional property that the subgraph of G induced by the set of all vertices of positive weight under f has no isolated vertex. The total Roman domination number $\gamma_{\text{tR}}(G)$ is the minimum weight of a TRDF on G . A TRDF with minimum weight $\gamma_{\text{tR}}(G)$ is called a $\gamma_{\text{tR}}(G)$ -function. The concept of total Roman domination in graphs was introduced by Liu and Chang [7] and has been studied in [1, 3].

A TRDF $f : V(G) \rightarrow \{0, 1, 2\}$ is called a *global total Roman dominating function* (GTRDF) if f is also a TRDF of the complement $\overline{G}$ of G . The *global total Roman*

domination number of G , denoted by $\gamma_{\text{gtR}}(G)$, is the minimum weight of a GTRDF on G . A $\gamma_{\text{gtR}}(G)$ -function is a GTRDF of G with weight $\gamma_{\text{gtR}}(G)$. Since $V_1 \cup V_2$ is a global total dominating set when $f = (V_0, V_1, V_2)$ is a GTRDF, and since placing weight 2 at the vertices of a global total dominating set yields a GTRDF, we observe that

$$\gamma_{\text{gt}}(G) \leq \gamma_{\text{gtR}}(G) \leq 2\gamma_{\text{gt}}(G). \quad (1)$$

On the other hand, since every GTRDF of G is a TRDF of G and $\overline{G}$, we conclude that

$$\gamma_{\text{gtR}}(G) \geq \max\{\gamma_{\text{tR}}(G), \gamma_{\text{tR}}(\overline{G})\}. \quad (2)$$

Our aim in this paper is to initiate the study of the global total Roman domination numbers in graphs. In particular, we relate the global total Roman domination and the total Roman domination and the global Roman domination number.

We make use of the following results in this paper.

Observation 1. If G is a graph of order n , then

$$\gamma_{\text{gtR}}(G) \leq n.$$

Let $\mathcal{G}$ be the family of graphs that can be obtained from a 4-cycle $(v_1 v_2 v_3 v_4)$ by adding $k_1 + k_2 \geq 1$ vertex-disjoint paths P_2 and joining v_1 to the end of k_1 such paths and joining v_2 to the end of k_2 such paths (possibly, $k_1 = 0$ or $k_2 = 0$). Let $\mathcal{H}$ be the family of graphs that can be obtained from a double star by subdividing each pendant edge once and subdividing the non-pendant edge $r \geq 0$ times.

Theorem A ([1]). Let G be a connected graph of order n . Then $\gamma_{\text{tR}}(G) = n$ if and only if one of the following holds.

1. G is a path or a cycle.
2. G is a corona, $\text{cor}(F)$, of some graph F .
3. G is a subdivided star.
4. $G \in \mathcal{G} \cup \mathcal{H}$.

Theorem B ([6]). Let G be a graph of order n such that neither G nor $\overline{G}$ have an isolated vertex. Then $\gamma_{\text{gt}}(G) = n$ if and only if $G = P_4$ or mK_2 or $\overline{mK_2}$; $m \geq 2$.

2. Basic Properties

In this section, we study some basic properties of the global total Roman domination number. First, we characterize all graphs G with the property that $\gamma_{\text{gtR}}(G) = n$.

Proposition 2. Let G be a graph of order n . Then $\gamma_{\text{gtR}}(G) = n$ if and only if for every set S of vertices with $|S| \geq 2$ and each subset B of $N_G(S) - S$ with $|B| > |S|$ and $\delta(G[V - B]) > 0$ and $\delta(\overline{G}[V - B]) > 0$, there is a vertex $x \in B$ such that $S \subseteq N_G(x)$.

Proof. Let for every set S of vertices with $|S| \geq 2$ and each subset B of $N_G(S) - S$ with $|B| > |S|$ and $\delta(G[V - B]) > 0$ and $\delta(\overline{G}[V - B]) > 0$, there exist a vertex $x \in B$ such that $S \subseteq N_G(x)$. Assume, to the contrary, that $\gamma_{\text{gtR}}(G) < n$, and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{gtR}}(G)$ -function. Since f is a $\gamma_{\text{gtR}}(G)$ -function, we have $|V_2| \geq 2$, $V_0 \subseteq N(V_2) - V_2$ and $\delta(G[V_1 \cup V_2]) > 0$ and $\delta(\overline{G}[V_1 \cup V_2]) > 0$. On the other hand, it follows from $\gamma_{\text{gtR}}(G) < n$ that $|V_0| > |V_2|$. Let $S = V_2$ and $B = V_0$. Since $V_0 \subseteq N(V_2) - V_2$, we deduce that $V_2 \not\subseteq N_G(x)$ for each $x \in V_0$, a contradiction.

Conversely, let $\gamma_{\text{gtR}}(G) = n$. Suppose S is an arbitrary set of vertices with $|S| \geq 2$ and B a subset of $N_G(S) - S$ with $|B| > |S|$ and $\delta(G[V - B]) > 0$ and $\delta(\overline{G}[V - B]) > 0$. Assume, to the contrary, that $S \not\subseteq N_G(x)$ for each $x \in B$. Then each vertex $x \in B$ is adjacent to some vertex of S in $\overline{G}$. Hence $B \subseteq N_{\overline{G}}(S)$. Now, it is easy to see that the function $f = (B, V(G) - (S \cup B), S)$ is a GTRDF of G and so $\gamma_{\text{gtR}}(G) < n$ which is a contradiction. Thus, there exists a vertex $x \in B$ so that $S \subseteq N_G(x)$ and the proof is complete. $\square$

Proposition 3. *For any graph G of order $n \geq 6$,*

$$\gamma_{\text{gtR}}(G) \geq 6$$

with equality if and only if $n = 6$ or there exists a 4-subset $S = \{x, y, u, v\} \subseteq V(G)$ such that $N(u) \cap N(v) \cap (V(G) \setminus S) = \emptyset$, $(V(G) \setminus S) \subseteq N(u) \cup N(v)$ and both of $G[S]$ and $\overline{G}[S]$ have no isolated vertex.

Proof. Assume, to the contrary, that $\gamma_{\text{gtR}}(G) \leq 5$ and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{gtR}}(G)$ -function. Then clearly $|V_2| \leq 2$. On the other hand, since $n \geq 6$ and $\gamma_{\text{gtR}}(G) \leq 5$, we must have $|V_2| \geq 1$ and $V_0 \neq \emptyset$. Since f is a GTRDF of $\overline{G}$, we conclude that $|V_2| = 2$. Hence $|V_1| \leq 1$. But then either $G[V_1 \cup V_2]$ or $\overline{G}[V_1 \cup V_2]$ has an isolated vertex which is a contradiction. Thus, $\gamma_{\text{gtR}}(G) \geq 6$.

If $n = 6$, then the result is immediate by Observation 1. If there exists a 4-subset $S = \{x, y, u, v\} \subseteq V(G)$ such that $N(u) \cap N(v) \cap (V(G) \setminus S) = \emptyset$, $(V(G) \setminus S) \subseteq N(u) \cup N(v)$ and $G[S]$ and $\overline{G}[S]$ have no isolated vertex, then the function $f : V(G) \rightarrow \{0, 1, 2\}$ defined by $f(u) = f(v) = 2$, $f(x) = f(y) = 1$ and $f(z) = 0$ otherwise, is a GTRDF of G with weight 6 and so $\gamma_{\text{gtR}}(G) = 6$.

Conversely, let $\gamma_{\text{gtR}}(G) = 6$. If $n = 6$, then we are done. Let $n \geq 7$ and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{gtR}}(G)$ -function. Then we have $|V_2| \leq 3$ and $V_0 \neq \emptyset$. If $|V_2| = 3$, then $|V_1| = 0$ and either $G[V_2]$ or $\overline{G}[V_2]$ has an isolated vertex that leads to a contradiction. Hence, we may assume that $|V_2| = 2$ implying that $|V_1| = 2$. Let $V_1 = \{x, y\}$, $V_2 = \{u, v\}$ and $S = \{x, y, u, v\}$. Since f is a GTRDF of G and $\overline{G}$, we deduce that $N(u) \cap N(v) \cap (V(G) \setminus S) = \emptyset$, $(V(G) \setminus S) \subseteq N(u) \cup N(v)$, and both of $G[S]$ and $\overline{G}[S]$ have no isolated vertex. This completes the proof. $\square$

Next, we characterize extreme graphs for both bounds in (1).

Proposition 4. *Let G be any graph. Then $\gamma_{\text{gtR}}(G) = \gamma_{\text{gt}}(G)$ if and only if $G = P_4$ or mK_2 or $\overline{mK_2}$; $m \geq 2$.*

Proof. One side is clear by (1) and Theorem B. Let $\gamma_{\text{gtR}}(G) = \gamma_{\text{gt}}(G)$ and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{gtR}}(G)$ -function. Since $V_1 \cup V_2$ is a global total dominating set of G we have

$$\gamma_{\text{gt}}(G) \leq |V_1| + |V_2| \leq |V_1| + 2|V_2| = \gamma_{\text{gtR}}(G).$$

It follows that $|V_2| = 0$ and so $|V_0| = 0$. Thus, $\gamma_{\text{gt}}(G) = n$ and the result follows from Theorem B. $\square$

Proposition 5. *Let G be a graph. Then, $\gamma_{\text{gtR}}(G) = 2\gamma_{\text{gt}}(G)$ if and only if there exists a $\gamma_{\text{gtR}}(G)$ -function $f = (V_0, V_1, V_2)$ such that $V_1 = \emptyset$.*

Proof. Let $\gamma_{\text{gtR}}(G) = 2\gamma_{\text{gt}}(G)$ and let S be a $\gamma_{\text{gt}}(G)$ -set. The function $f = (V \setminus S, \emptyset, S)$ is a GTRDF on G , and so $2\gamma_{\text{gt}}(G) = \gamma_{\text{gtR}}(G) \leq f(V(G)) = 2|S| = 2\gamma_{\text{gt}}(G)$. It follows that $\gamma_{\text{gtR}}(G) = f(V(G))$ and hence f is a $\gamma_{\text{gtR}}(G)$ -function satisfying $V_1 = \emptyset$.

Conversely, let there exist a $\gamma_{\text{gtR}}(G)$ -function $f = (V_0, V_1, V_2)$ satisfying $V_1 = \emptyset$. Since the set $V_1 \cup V_2 = V_2$ is a GTDS of G , we have $\gamma_{\text{gt}}(G) \leq |V_2| = \frac{1}{2}\gamma_{\text{gtR}}(G)$. Now, the result follows from (1). $\square$

3. Global Total Roman Domination Versus Total Roman Domination

In this section, we relate the global total Roman domination and the total Roman domination number. We observe that for any graph G , $\gamma_{\text{tR}}(G) \leq \gamma_{\text{gtR}}(G)$. First, we provide some sufficient conditions to have $\gamma_{\text{gtR}}(G) = \gamma_{\text{tR}}(G)$.

Proposition 6. *If G is a disconnected graph with at least two components of order at least three, then*

$$\gamma_{\text{gtR}}(G) = \gamma_{\text{tR}}(G).$$

Proof. Let $G_1, G_2, \dots, G_k$ be the components of G , and assume, without loss of generality, that $|V(G_i)| \geq 3$ for $i = 1, 2$. Let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(G)$ -function such that $|V_2|$ is maximum. Clearly, there exists a vertex $u_i \in V(G_i) \cap V_2$ such that $\sum_{v \in N(u_i)} f(v) > 0$ for $i = 1, 2$. This implies that f is a GTRDF of G and so $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G)$. Now, the result is immediate by (2). $\square$

Theorem 7. *If G is a connected graph with radius $r(G) \geq 5$, then $\gamma_{\text{gtR}}(G) = \gamma_{\text{tR}}(G)$.*

Proof. Assume $f = (V_0, V_1, V_2)$ is a $\gamma_{\text{tR}}(G)$ -function such that $|V_1|$ is minimum. We show that f is a GTRDF of G . Suppose, to the contrary, that f is not a TRDF of $\overline{G}$. Then there exists a vertex $v \in V_1 \cup V_2$ such that $(V_1 \cup V_2) - \{v\} \subseteq N_G(v)$, or there exists a vertex $v \in V_0$ such that $V_2 \subseteq N(v)$. In the first case, we have $d(v, u) \leq 2$ for each $u \in V(G)$ and hence $r(G) \leq 2$, which is a contradiction. Let there exist a vertex

$v \in V_0$ such that $V_2 \subseteq N(v)$. Suppose u is an arbitrary vertex in $V(G)$. If $u \in V_2$, then $d(u, v) = 1$. If $u \in V_0$, then u and v have a common neighbor in V_2 and hence $d(u, v) \leq 2$. Let $u \in V_1$. If there exists a vertex $w \in V_0 \cup V_2$ such that $d(u, w) \leq 2$, then clearly $d(u, v) \leq 4$. Hence, we may assume that every vertex at distance at most two from u in G , is in V_1 . Since G is connected, there exists a path uu_1u_2 in G for which $f(u) = f(u_1) = f(u_2) = 1$. If the induced subgraph $G[V_1 \cup V_2] - \{u\}$ has no isolated vertex, then the function $g = (V_0 \cup \{u\}, V_1 - \{u, u_1\}, V_2 \cup \{u_1\})$ is a $\gamma_{\text{tR}}(G)$ -function which contradicts the choice of f . If the subgraph $G[V_1 \cup V_2] - \{u\}$ has an isolated vertex, say w , then the function $g = (V_0 \cup \{w\}, V_1 - \{u, w\}, V_2 \cup \{u\})$ is a $\gamma_{\text{tR}}(G)$ -function which leads to a contradiction again. Therefore, $d(u, v) \leq 4$ for each $u \in V(G) - \{v\}$, a contradiction. Thus, f is a TRDF of $\overline{G}$ and this completes the proof. $\square$

Proposition 8. *If the graph G has a component isomorphic to K_2 , then*

$$\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 1.$$

Proof. Let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(G)$ -function. If u, v are two vertices of the component isomorphic to K_2 , then clearly $u, v \in V_1$. Now define $g : V(G) \rightarrow \{0, 1, 2\}$ by $g(u) = 2$ and $g(x) = f(x)$ for $x \in V(G) - \{u\}$. Clearly, g is a GTRDF of G with weight $\gamma_{\text{tR}}(G) + 1$ and so $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 1$. $\square$

Let H be the disjoint union of rK_2 and a complete graph K_s with $s \geq 4$. Then $\gamma_{\text{tR}}(H) = 2r + 3$ and $\gamma_{\text{gtR}}(H) = 2r + 4$. This example demonstrates that Proposition 8 is tight.

Proposition 9. *If G is a connected triangle-free graph of order $n \geq 4$, then*

$$\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 2.$$

Proof. If $\gamma_{\text{tR}}(G) \geq n - 2$, then the result is immediate. Let $\gamma_{\text{tR}}(G) \leq n - 3$ and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(G)$ -function. Clearly, $|V_2| \geq 1$. First, let $|V_2| = 1$ and let $V_2 = \{u\}$. Then $V_0 \subseteq N(u)$. Since f is a total Roman dominating function of G and since $\delta(\overline{G}) > 0$, there are two vertices $v, w \in V_1$ such that $uv \in E(G)$ and $uw \in E(\overline{G})$. Also, we deduce from $\delta(\overline{G}) > 0$ that there exists a vertex $z \in V(G)$ such that $vz \in E(\overline{G})$. Define $g : V(G) \rightarrow \{0, 1, 2\}$ by $g(v) = 2, g(z) = 1$ and $g(y) = f(y)$ for $y \in V(G) - \{v, z\}$. Since G is triangle-free, we have $V_0^g \subseteq N_{\overline{G}}(v)$ and that $V_1^g \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(v)$. On the other hand, neither u nor v is an isolated vertex in $\overline{G}$ and hence the subgraph of $\overline{G}$ induced by $V_1^g \cup V_2^g$ has no isolated vertex. Thus, g is a GTRDF of G with weight $\gamma_{\text{tR}}(G) + 2$ and so $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 2$.

Now, let $|V_2| \geq 2$. If there exist two adjacent vertices $u, v \in V_2$, then let $u' \in N(u) \cap V_0$ and $v' \in N(v) \cap V_0$ and define $g : V(G) \rightarrow \{0, 1, 2\}$ by $g(u') = g(v') = 1$ and $g(w) = f(w)$ for $w \in V(G) - \{u', v'\}$. Since G is triangle-free, we have $V_0^g = V_0 - \{u', v'\} \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(v)$, $V_1^g - \{u', v'\} \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(v)$ and $V_2^g - \{u, v\} \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(v)$. On the other hand, we have $uv' \in E(\overline{G})$ and $vu' \in E(\overline{G})$. Thus, g is a GTRDF of G with weight $\gamma_{\text{tR}}(G) + 2$ yielding $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 2$. Hence,

we may assume that $G[V_2]$ is an independent set. Let $u, v \in V_2$. Since $G[V_1 \cup V_2]$ has no isolated vertex, u has at least one neighbor in V_1 , say u_1 . Let u_2 be a private neighbor of u in V_0 with respect to V_2 , and define $g : V(G) \rightarrow \{0, 1, 2\}$ by $g(u_1) = 2, g(u_2) = 1$ and $g(w) = f(w)$ for $w \in V(G) - \{u_1, u_2\}$. Since G is triangle-free, we have $V_0^g \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(u_1)$ and $(V_1^g \cup V_2^g) - \{u, u_1\} \subseteq N_{\overline{G}}(u) \cup N_{\overline{G}}(u_1)$. Moreover, we have $uv \in E(\overline{G})$ and $u_1 u_2 \in E(\overline{G})$. Thus, g is a GTRDF of G with weight $\gamma_{\text{tR}}(G) + 2$ and so $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 2$. $\square$

Proposition 10. *Let G be a graph. If $\text{diam}(G) \geq 5$, then*

$$\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 1.$$

Proof. If G has a component isomorphic to K_2 , then the result follows from Proposition 8. If G is a disconnected graph with no component isomorphic to K_2 , then we have $\gamma_{\text{gtR}}(G) = \gamma_{\text{tR}}(G)$ by Proposition 6. Henceforth, we assume that G is connected. Let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(G)$ -function such that $|V_2|$ is maximum. If f is a TRDF of $\overline{G}$, then we are done. Suppose that f is not a TRDF of $\overline{G}$. Then there exists a vertex $v \in V_0$ such that $V_2 \subseteq N(v)$ or there exists a vertex $v' \in V_1 \cup V_2$ such that v' is adjacent to every vertex in $V_1 \cup V_2$. If there exists a vertex $v' \in V_1 \cup V_2$ such that v' is adjacent to every vertex in $V_1 \cup V_2$, then we have $d(v', u) \leq 2$ for each $u \in V(G)$ and hence $\text{diam}(G) \leq 4$ which is a contradiction. Thus, we may assume that $\overline{G}[V_1 \cup V_2]$ has no isolated vertex. Let there exist a vertex $v \in V_0$ such that $V_2 \subseteq N(v)$. This implies that $d(x, y) \leq 2$ for every pair of vertices $x, y \in V_2$, $d(x, y) \leq 3$ for every pair of vertices x, y where $x \in V_0, y \in V_2$ and $d(x, y) \leq 4$ for every pair of vertices $x, y \in V_0$. We claim that there are two vertices $x \in V_1$ and $y \in V_2$ such that $d(x, y) \geq 3$. In this case, since no vertex in G is adjacent to both x, y , the function $g : V \rightarrow \{0, 1, 2\}$ defined by $g(x) = 2$ and $g(z) = f(z)$ for $z \in V(G) - \{x\}$, is a GTRDF of G and so $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 1$ as desired. Assume, to the contrary, that $d(x, y) \leq 2$ for every pair of vertices x, y where $x \in V_1$ and $y \in V_2$. This implies that $d(w, z) \leq 3$ for every pair of vertices w, z where $w \in V_1$ and $z \in V_0$ and $d(w, z) \leq 4$ for every pair of vertices $w, z \in V_1$ which contradicts the hypothesis $\text{diam}(G) \geq 5$. This completes the proof. $\square$

Proposition 11. *Let G be a connected graph with diameter three or four. Then*

$$\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + \text{diam}(G) + (-1)^{\text{diam}(G)+1}. \quad (1)$$

Proof. Let f be a $\gamma_{\text{tR}}(G)$ -function, and let u and v be two vertices of G at distance $\text{diam}(G)$. If $\text{diam}(G) = 4$, then $(V_1 \cup V_2) \cap N[u] \neq \emptyset$. Let u_1 be a neighbor of u in $V_1 \cup V_2$. Then no vertex in G is adjacent to both u_1 and v . Hence, the function $g : V \rightarrow \{0, 1, 2\}$ defined by $g(u_1) = g(v) = 2$ and $g(x) = f(x)$ for $x \in V(G) - \{u_1, v\}$, is a GTRDF of G and therefore $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 3$.

If $\text{diam}(G) = 3$, then no vertex in G is adjacent to both u and v . Therefore, the function $g : V \rightarrow \{0, 1, 2\}$ defined by $g(u) = g(v) = 2$ and $g(x) = f(x)$ for $x \in V(G) - \{u, v\}$, is a GTRDF of G and hence $\gamma_{\text{gtR}}(G) \leq \gamma_{\text{tR}}(G) + 4$. $\square$

Let $n \geq 6$ and let G be the graph obtained from the complete graph K_n with vertex set $\{x_1, \dots, x_n\}$ by deleting the edges $x_1x_2, x_2x_3, \dots, x_{n-1}x_n, x_nx_1$. Clearly, $\gamma_{tR}(G) = 4$ and $\gamma_{gtR}(G) = n$. Thus, the difference between $\gamma_{gtR}(G)$ and $\gamma_{tR}(G)$ may be very large when $\text{diam}(G) = 2$. So the case $\text{diam}(G) = \text{diam}(\overline{G}) = 2$ remained open.

3.1. Global total Roman domination numbers of trees

As a special case of (2) and Theorem 9, we observe that for any tree T of order $n \geq 4$, $\gamma_{tR}(T) \leq \gamma_{gtR}(T) \leq \gamma_{tR}(T) + 2$. In the sequel, we characterize all extremal trees. The first result is an immediate consequence of Theorem 7.

Corollary 12. *For any tree T of order $n \geq 4$ with $\text{diam}(T) \geq 9$, $\gamma_{gtR}(T) = \gamma_{tR}(T)$.*

Lemma 13. *Let T be a tree of order $n \geq 4$, f a $\gamma_{tR}(T)$ -function and $u, v \in V_2$.*

- (1) *If $d(u, v) \geq 3$, then $\gamma_{gtR}(T) = \gamma_{tR}(T)$.*
- (2) *If $d(u, v) = 2$, then $\gamma_{gtR}(T) \leq \gamma_{tR}(T) + 1$.*
- (3) *If $d(u, v) = 2$, the common neighbor of u, v , say w , has positive value and there exists a vertex $y \in V(T) - (N[u] \cup N[v] \cup N[w])$ with $f(y) \geq 1$, then $\gamma_{gtR}(T) = \gamma_{tR}(T)$.*
- (4) *If $d(u, v) = 1$ and $(V(G) - (N[u] \cup N[v])) \cap (V_1 \cup V_2) \neq \emptyset$ or both u and v have a neighbor in $V_1 \cup V_2$, then $\gamma_{gtR}(T) = \gamma_{tR}(T)$.*
- (5) *If $d(u, v) = 1$ and T is not a double star, then $\gamma_{gtR}(T) \leq \gamma_{tR}(T) + 1$.*

Proof. (1) Since T is a tree and $d(u, v) \geq 3$, we have $N_T(u) \cap N_T(v) = \emptyset$. Thus, f is a GTRDF of G and hence $\gamma_{gtR}(T) \leq \gamma_{tR}(T)$. Now, the result follows from (2).
 (2) Since T is a tree and $d(u, v) = 2$, u and v have exactly one common neighbor, say w . If $f(w) = 0$, then define $g : V(T) \rightarrow \{0, 1, 2\}$ by $g(w) = 1$ and $g(z) = f(z)$ for $z \in V(T) - \{w\}$. If $f(w) > 0$, then let $u' \in V_0$ be a private neighbor of u with respect to V_2 , and define $g : V(T) \rightarrow \{0, 1, 2\}$ by $g(u') = 1$ and $g(z) = f(z)$ for $z \in V(T) - \{u'\}$. It is easy to see that g is a GTRDF of T and hence $\gamma_{gtR}(T) \leq \gamma_{tR}(T) + 1$.
 (3) It is trivial.
 (4) It is trivial.
 (5) Since T is not a double star, there is a vertex in $(V_1 \cup V_2) \setminus \{u, v\}$, say x . If $d(x, \{u, v\}) \geq 2$, then $\gamma_{gtR}(T) = \gamma_{tR}(T)$ by part 4. Let $d(x, \{u, v\}) = 1$ and assume, without loss of generality, that $d(x, u) = 1$. Assume $y \in V_0 \cap N(v)$ and define $g : V(T) \rightarrow \{0, 1, 2\}$ by $g(y) = 1$ and $g(z) = f(z)$ for $z \in V(T) - \{y\}$. Clearly, g is a GTRDF of G of weight $\gamma_{tR}(T) + 1$ and so $\gamma_{gtR}(T) \leq \gamma_{tR}(T) + 1$. $\square$

Next results are immediate consequences of Lemma 13.

Corollary 14. *Let T be a tree of order $n \geq 4$. If T has two strong support vertices at distance at least three from each other, then $\gamma_{gtR}(T) = \gamma_{tR}(T)$.*

Corollary 15. *If T is a tree of order $n \geq 4$ different from double stars with $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T) + 2$ and $f = (V_0, V_1, V_2)$ is a $\gamma_{\text{tR}}(T)$ -function, then $|V_2| = 1$.*

Theorem 16. *Let T be a tree of order $n \geq 5$ with $\text{diam}(T) \geq 4$. Then $\gamma_{\text{gtR}}(T) \leq \gamma_{\text{tR}}(T) + 1$.*

Proof. By Proposition 10, we may assume that $\text{diam}(T) = 4$. Assume, to the contrary, that $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T) + 2$ and let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(T)$ -function. We conclude from $\gamma_{\text{tR}}(T) = \gamma_{\text{gtR}}(T) - 2 \leq n - 2$ and Corollary 15 that $|V_2| = 1$. Let $P := u_1 u_2 u_3 u_4 u_5$ be a diametral path in T . Since u_1 is a leaf, $f(u_1) \neq 2$. we consider two cases.

Case 1. $f(u_1) = 0$.

Then $f(u_2) = 2$. Since $|V_2| = 1$, we deduce that $f(u_4) = f(u_5) = 1$. Then clearly the function $g : V(T) \rightarrow \{0, 1, 2\}$ defined by $g(u_5) = 2$ and $g(x) = f(x)$ for $x \in V(T) - \{u_5\}$, is a GTRDF of T of weight $\gamma_{\text{tR}}(T) + 1$ which leads to a contradiction.

Case 2. $f(u_1) = 1$.

Then $f(u_2) > 0$. If $f(u_2) = 2$, then we get a contradiction as above. Assume that $f(u_2) = 1$. If $f(u_3) > 0$, then $g = (V_0 \cup \{u_1\}, V_1 - \{u_1\}, V_2 \cup \{u_2\})$ is a TRDF of G with $|V_2^g| = 2$ which contradicts Corollary 15. If $f(u_3) = 0$, then u_3 has a neighbor in V_2 and the function $g = (V_0, V_1 - \{u_1\}, V_2 \cup \{u_1\})$ is a GTRDF of G of weight $\gamma_{\text{tR}}(T) + 1$ which is a contradiction. This completes the proof. $\square$

Corollary 17. *Let T be a tree. Then $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T) + 2$ if and only if $T = DS(r, s)$ for integers $r \geq s \geq 2$.*

In what follows, we characterize all trees T with the property that $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T)$.

Lemma 18. *If T is a tree with $\text{diam}(T) \geq 5$, then $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T)$.*

Proof. Suppose, to the contrary, that $\gamma_{\text{gtR}}(T) \geq \gamma_{\text{tR}}(T) + 1$. It follows from Theorem 16 that $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T) + 1$. Let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{tR}}(T)$ -function such that $|V_2|$ is maximum and let $P := u_0 u_1 u_2 \dots u_{d-2} u_{d-1} u_d$ be a diametral path in T . We consider two cases.

Case 1. $\deg(u_1) \geq 3$ (the case $\deg(u_{d-1}) \geq 3$ is similar).

Then we may assume that $f(u_1) = 2$ and $f(u_2) > 0$. If $f(u_{d-1}) = 2$ or $\deg(u_{d-1}) \geq 3$, then we get a contradiction by Lemma 13 (part 1) or Corollary 14. Henceforth, we assume that $\deg(u_{d-1}) = 2$ and $f(u_{d-1}) = f(u_d) = 1$. Using a similar argument, we may assume that each support vertex adjacent to u_{d-2} has degree 2. If $f(u_{d-2}) > 0$, then $g = (V_0 \cup \{u_d\}, V_1 - \{u_{d-1}, u_d\}, V_2 \cup \{u_{d-1}\})$ is a TRDF of G which contradicts the choice of f . Hence $f(u_{d-2}) = 0$ and so $f(u_{d-3}) = 2$. But this leads to a contradiction by Lemma 13 (parts 1–4).

Case 2. $\deg(u_1) = \deg(u_{d-1}) = 2$.

If $f(u_1) = 2$ or $f(u_{d-1}) = 2$, then the result follows as Case 1. Thus, we may assume that $f(u_0) = f(u_1) = f(u_{d-1}) = f(u_d) = 1$. Since $|V_2|$ is maximum, we must have $f(u_2) = f(u_{d-2}) = 0$. To dominate u_2 and u_{d-2} , we must have $V_2 \cap N(u_2) \neq \emptyset$ and $V_2 \cap N(u_{d-2}) \neq \emptyset$. Let $x \in V_2 \cap N(u_2)$ and $y \in V_2 \cap N(u_{d-2})$ such that $d(x, y)$ is as small as possible. If $d(x, y) \geq 3$, then we obtain a contradiction by Lemma 13 (part 1). Assume that $d(x, y) \leq 2$. This implies that $6 \leq d \leq 8$. If $\text{diam}(T) = 7$, then we must have $x = u_3$ and $y = u_4$ that leads to a contradiction by Lemma 13 (part 4). Consider two subcases.

Subcase 2.1. $\text{diam}(T) = 8$.

It follows from $d(x, y) \leq 2$ that $x = u_3$ and $y = u_5$. If $f(u_4) \geq 1$, then we get a contradiction by Lemma 13 (part 3). Let $f(u_4) = 0$. To totally dominate u_3 , u_3 must have a neighbor z_1 in $V_1 \cup V_2$. By assumption and Lemma 13 (part 1), we must have $z_1 \in V_1$. If z_1 is a leaf, then the function g defined by $g(z_1) = g(u_0) = 0, g(u_1) = 2, g(u_2) = 1$ and $g(u) = f(u)$ otherwise, is a $\gamma_{\text{tR}}(T)$ -function which contradicts the choice of f . If z_1 is a support vertex and z_2 is a leaf adjacent to z_1 , then clearly $f(z_2) = 1$ and the function $g = (V_0 \cup \{z_2\}, V_1 - \{z_1, z_2\}, V_2 \cup \{z_1\})$ is a $\gamma_{\text{tR}}(T)$ -function which contradicts the choice of f again. Thus, there exists a path $u_4 z_1 z_2 z_3$ in T . As above, we may assume that $\deg(z_2) = 2$ and $f(z_2) = f(z_3) = 1$. Then the function $g = (V_0 \cup \{z_3\}, V_1 - \{z_2, z_3\}, V_2 \cup \{z_2\})$ is a $\gamma_{\text{tR}}(T)$ -function and we get a contradiction by Lemma 13 (part 1).

Subcase 2.2. $\text{diam}(T) = 6$.

Since $d(x, y) \leq 2$ and since $d(x, y)$ is as small as possible, we must have $x = y = u_3$. As Subcase 2.1, we may assume that u_3 has a neighbor z_1 in V_1 . Using an argument similar to that described in Subcase 2.1, we obtain a contradiction and this completes the proof. $\square$

Let $\mathcal{F}_1$ be the family of trees T obtained from stars $K_{1,r_1}, K_{1,r_2}, \dots, K_{1,r_s}$, where $s \geq 3$ and $r_i = 2$ for some i , by adding a vertex and joining it to the center of stars. Also, let $\mathcal{F}_2$ be the family of trees T obtained from $\overline{K_t}$ ($t \geq 1$) and stars $K_{1,r_1}, K_{1,r_2}, \dots, K_{1,r_s}$, where $s \geq 2$ and $r_i = 2$ for some i , by adding a vertex and joining it to the vertices of $\overline{K_t}$ and to the center of stars.

Lemma 19. *Let T be a tree with $\text{diam}(T) = 4$. Then $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T)$ if and only if $T \in \mathcal{F}_1 \cup \mathcal{F}_2 \cup \{P_5\}$.*

Proof. One side is clear. Let $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T)$. Assume that $f = (V_0, V_1, V_2)$ is a $\gamma_{\text{tR}}(T)$ -function such that $|V_2|$ is maximum and $P := u_1 u_2 u_3 u_4 u_5$ is a diametral path in T . If $T = P_5$, then we are done. Let $T \neq P_5$. If $\deg(u_3) = 2$, then clearly $\gamma_{\text{tR}}(T) = 5$ which leads to a contradiction by Proposition 3. Hence assume that $\deg(u_3) \geq 3$. First, let u_3 is not a support vertex. If each support vertex adjacent

to u_3 has degree at least three, then T has a unique $\gamma_{\text{tR}}(T)$ -function that assigns 2 to each support vertex, 1 to u_3 and 0 to the remaining vertices. But this is not a GTRDF of T and so $\gamma_{\text{gtR}}(T) > \gamma_{\text{tR}}(T)$ which is a contradiction. Thus, u_3 is adjacent to a support vertex of degree 2 and so $T \in \mathcal{F}_1$.

Now, let u_3 be a support vertex and u a leaf adjacent to u_3 . If each support vertex adjacent to u_3 has degree at least three, then either f must assign 2 to the support vertices adjacent to u_3 , 2 to u_3 , and 0 to the remaining vertices or f must assign 2 to the support vertices adjacent to u_3 , 1 to u_3, u , and 0 to the remaining vertices. But then f is not a GTRDF of T and this implies that $\gamma_{\text{gtR}}(T) > \gamma_{\text{tR}}(T)$, a contradiction. Hence u_3 is adjacent to a support vertex of degree 2 and so $T \in \mathcal{F}_2$. $\square$

Now, we are ready to state the main theorem of this section.

Theorem 20. *Let T be a tree. Then $\gamma_{\text{gtR}}(T) = \gamma_{\text{tR}}(T)$ if and only if $\text{diam}(T) \geq 5$ or $T \in \mathcal{F}_1 \cup \mathcal{F}_2 \cup \{P_5\}$.*

4. Global Total Roman Domination Versus Global Roman Domination

In this section, we relate the global total Roman domination and the global Roman domination number. Since every global total Roman dominating function of a graph G is a global Roman dominating function of G , we have $\gamma_{\text{gR}}(G) \leq \gamma_{\text{gtR}}(G)$. We establish next an upper bound on the global total Roman domination number in terms of the global Roman domination number.

Theorem 21. *Let G be a graph of order $n \geq 4$. Then $\gamma_{\text{gtR}}(G) \leq 2(\gamma_{\text{gR}}(G) - 1)$.*

Proof. Let $f = (V_0, V_1, V_2)$ be a $\gamma_{\text{gR}}(G)$ -function such that $|V_2|$ is maximum. If $V_2 = \emptyset$, then $\gamma_{\text{gR}}(G) = n$ and so $\gamma_{\text{gtR}}(G) = \gamma_{\text{gR}}(G) < 2(\gamma_{\text{gR}}(G) - 1)$ since $\gamma_{\text{gR}}(G) \geq 4$. Hence let $V_2 \neq \emptyset$. Consider two cases.

Case 1. $V_1 \neq \emptyset$.

If there exists a vertex $u \in V_1$ such that $N(u) \cap V_2 \neq \emptyset$ and $V_2 \not\subseteq N(u)$, then the function $g = (V_0 \cup \{u\}, V_1 - \{u\}, V_2)$ is a GRDF of G of weight less than $f(V(G))$, a contradiction. Thus for each $u \in V_1$, either $N(u) \cap V_2 = \emptyset$ or $V_2 \subseteq N(u)$. Let $V'_1 = \{u \in V_1 \mid N(u) \cap V_2 = \emptyset\}$ and $V''_1 = \{u \in V_1 \mid V_2 \subseteq N(u)\}$. If there is an edge $uv \in E(G)$ where $u \in V'_1$ and $v \in V''_1$, then the function $g = (V_0 \cup \{u\}, V_1 - \{u, v\}, V_2 \cup \{v\})$ is a GRDF of G of weight $\gamma_{\text{gR}}(G)$ which contradicts the choice of f . Hence there is no edge between V'_1 and V''_1 in G . If there is an edge $uv \in E(G)$ where $u, v \in V'_1$, then the function $g = (V_0 \cup \{v\}, V_1 - \{u, v\}, V_2 \cup \{u\})$ is clearly a GRDF of G of weight $\gamma_{\text{gR}}(G)$ which contradicts the choice of f again. Thus, V'_1 is an independent set. We now show that V'_1 is a packing in G . Suppose, to the contrary, that u and v are two distinct vertices in V'_1 at distance 2 apart, and let w

be a common neighbor of u and v . Since V_1' is an independent set in G , there is no edge between V_1' and V_1'' , and since neither u nor v are dominated by V_2 , we have $w \in V_0$. But then the function $g = (V_0 \cup \{u, v\} - \{w\}, V_1 - \{u, v\}, V_2 \cup \{w\})$ is a GRDF of G of weight $\gamma_{\text{gR}}(G)$ which once again a contradiction. Therefore, V_1' is a packing in G . Thus, all neighbors of a vertex in V_1' belong to V_0 . We consider two subcases.

Subcase 1.1. $V_1' \neq \emptyset$.

For each vertex $u \in V_1'$, we select an arbitrary neighbor, u' , of u and we let $W = \cup_{u \in V_1'} \{u'\}$. Note that $|W| = |V_1'| \leq |V_1|$ and $W \subseteq V_0$. Further, each vertex in W is adjacent to at least one vertex in V_2 in G and $\overline{G}$ and no vertex in V_1' is dominated by V_2 , implying that $G[V_1 \cup V_2 \cup W]$ has not isolated vertices in $\overline{G}$ and $|V_2| \geq 2$. If $V_1 \cup V_2 \cup W$ has no isolated vertices in G , then $g = (V_0 \setminus W, V_1 \cup W, V_2)$ is a GTRDF of G . Hence

$$\gamma_{\text{gR}}(G) \leq g(V(G)) = 2|V_1| + 2|V_2| \leq 2\gamma_{\text{gR}}(G) - 2|V_2| \leq 2(\gamma_{\text{gR}}(G) - 2).$$

Let $G[V_1 \cup V_2 \cup W]$ has isolated vertices in G and U is the set of isolated vertices in $G[V_1 \cup V_2 \cup W]$. Then we must have $V_1'' = \emptyset$. Note that $G[V_1' \cup W]$ contains no isolated vertex and so $U \subseteq V_2$. Further, each vertex in W is adjacent to at least one vertex in V_2 in G , implying that $U \subset V_2$. For each vertex $x \in U$, we select an arbitrary neighbor, x' , of x and we let $Z = \cup_{x \in U} \{x'\}$. Note that $|Z| \leq |U| < |V_2|$ and $Z \subseteq V_0 \setminus W$. Since each vertex in Z is adjacent to at least one vertex in V_2 in G and $\overline{G}$, $G[V_1 \cup V_2 \cup W \cup Z]$ has not isolated vertices in G and $\overline{G}$. Hence the function $g = (V_0 \setminus (Z \cup W), V_1 \cup (Z \cup W), V_2)$ is a GTRDF of G and this implies that

$$\gamma_{\text{gR}}(G) \leq g(V(G)) = 2|V_1| + 3|V_2| - 1 < 2|V_1| + 4|V_2| - 3 < 2(\gamma_{\text{gR}}(G) - 1).$$

Subcase 2. $V_1' = \emptyset$.

Then $V_1'' \neq \emptyset$. Since each vertex in V_1'' is adjacent to all vertices in V_2 and since $\overline{G}$ has no isolated vertex, we deduce that any vertex $x \in V_1''$ has a neighbor $x' \in V_0$ in $\overline{G}$. Let $Z = \{x' \in V_0 \mid x \in V_1''\}$. Then $G[V_1 \cup V_2 \cup Z]$ has isolated vertex in G and $\overline{G}$. Thus the function $g = (V_0 \setminus Z, V_1 \cup Z, V_2)$ is a GTRDF of G and so

$$\gamma_{\text{gR}}(G) \leq g(V(G)) = 2|V_1| + 2|V_2| \leq 2\gamma_{\text{gR}}(G) - 2|V_2| \leq 2(\gamma_{\text{gR}}(G) - 2).$$

Case 2. $V_1 = \emptyset$.

If the induced subgraphs $G[V_2]$ and $\overline{G}[V_2]$ have no isolated, then f is a GTRDF of G and this implies that $\gamma_{\text{gR}}(G) = \gamma_{\text{gR}}(G)$. Assume, without loss of generality, that $G[V_2]$ has an isolated vertex. Then clearly $\overline{G}[V_2]$ is connected. Let U be the set consisting of all isolated vertices of $G[V_2]$. For each vertex $u \in U$, we select an arbitrary neighbor, u' , of u and we let $W = \cup_{u \in U} \{u'\}$. Then $|W| \leq |V_2|$ and $W \subseteq V_0$. Since f is a GRDF, $G[V_2 \cup W]$ and $\overline{G}[V_2 \cup W]$ have no isolated vertex.

Hence the function $g = (V_0 \setminus W, W, V_2)$ is a GTRDF of G and so

$$\gamma_{\text{gtR}}(G) \leq g(V(G)) = 3|V_2| \leq 2\gamma_{\text{gR}}(G) - |V_2| \leq 2\gamma_{\text{gR}}(G) - 2 = 2(\gamma_{\text{gR}}(G) - 1). \quad \square$$

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